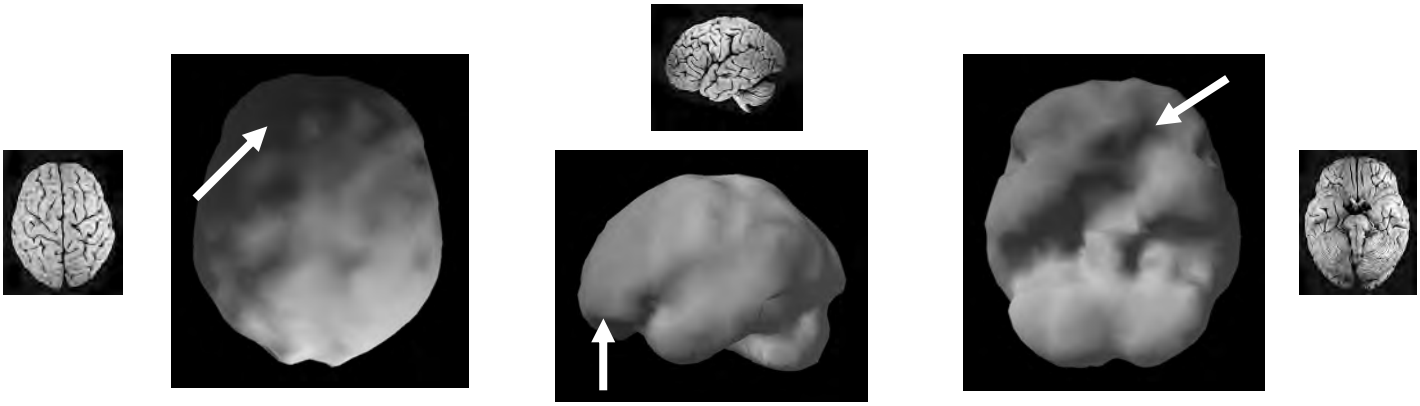
Amen Clinic Brain System Handout

Daniel G. Amen, MD, Amen Clinics, Inc.

The following handout is a compilation of the work done at the Amen Clinics. Please use this as a reference to understand the different brain systems as they relate to function, problems and treatments. We revise this handout as new information is available.

Prefrontal Cortex (PFC)



top down surface view | left side surface view | underside surface view

PFC Functions	PFC Problems
PFC Dorsal lateral	
Attention	Inattention
Planning	Lack of forethought
Follow through	Procrastination
PFC Inferior orbital	
Impulse control	Impulsive
Inhibition	Disinhibited
Judgment	Poor judgment
Empathy	Lack of empathy
Ethics	Lack of ethics
Morality	

Some Conditions Affecting the PFC

ADHD	Depression
Brain Trauma	Dementia
Schizophrenia	Antisocial PD
Conduct disorders	Borderline PD

PFC Behavioral Treatments

Coaching/goal setting	Organizational help
Intense aerobic exercise	Relationship counseling
Stimulating activities	Higher protein diet
Neurofeedback	HBOT, rTMS

PFC Supplements

Focus and Energy
Zinc (for hyperactivity)
Attention Support for Kids/Teens
Mood support – SAMe (SAMe Mood and Movement)
Fish oil – Omega 3 Power

PFC Meds

For ADD – stimulants, such as Adderall or Ritalin, or Stratterra
For Depression – Wellbutrin
For Low Energy – Provigil
For Psychosis -- Abilify

Positives w/low PFC activity: Spontaneous, risk taking, creative, "out of the box," curious, wide interests

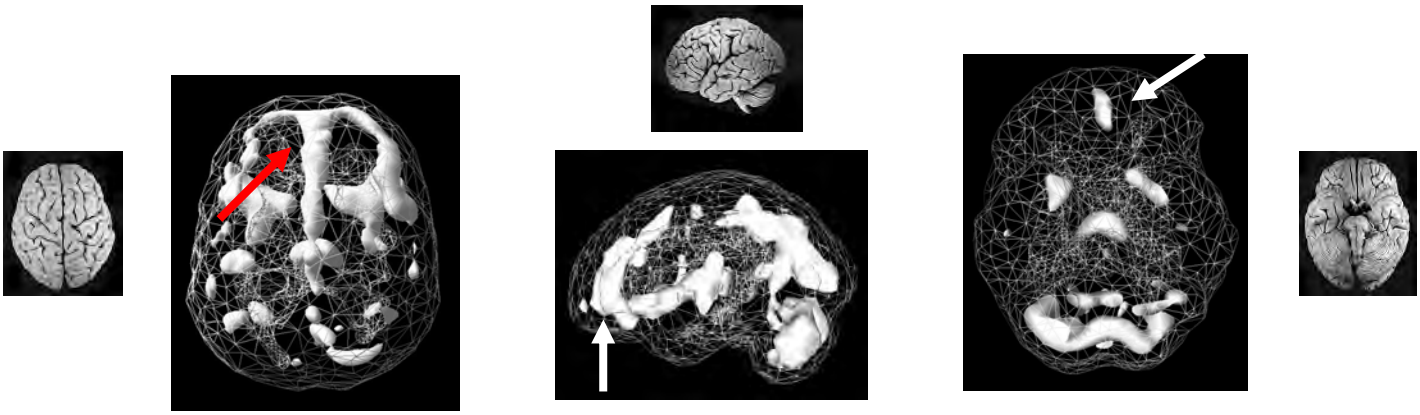
Decreases in the PFC during concentration is often associated with impulsivity, short attention span, distractibility and difficulties with organization and planning. We have seen a strong correlation between this finding and ADHD and ADD, especially when this occurs during the performance of a concentration task. When decreased activity in the PFC is seen during a resting state, it is often associated with depressive disorders, and may be responsive to antidepressant medication. When it is seen in both the resting and concentration states there may be a combination of depression and ADD or ADHD present. Clinical correlation is always needed. This pattern has also been seen in response to head injuries affecting this part of the brain, and later in life in some dementia processes.

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Anterior Cingulate Gyrus (ACG)



top down active view left side active view underside active view

ACG Functions
Brain's gear shifter
Cognitive flexibility
Cooperation
Go from idea to idea
See options
Go with the flow
Error detection

ACG Problems
Gets stuck, Trouble shifting
Inflexible, worries
Holds grudges, oppositional
Obsesses
Compulsions
Argumentative
Excessive error detection

Area 25 – mood states, activates GI system
Area 24v – emotional attention, communicates with limbic system
Area 24g (genu) -- attention to cold cognition
Area 24d – activated in nearly all cognitive tasks, focus to detail
Area 24d more posterior – error detection
Middle Cingulate – response to pain, also insular cortex
Posterior Cingulate – visual memory, visual attention, response to pain, posterior part has access to hippocampus and memory circuit

Some Conditions Affecting the ACG
OCD
Addictions
Eating Disorders
PTSD
Anxiety disorders
PMS
Chronic pain
Oppositional Defiant

ACG Behavioral Treatments
Distraction, Paradox, Options
Cognitive/behavioral strategies
Intense aerobic exercise
Lower protein/complex carbs diet

ACG Supplements
Worry and Insomnia – Serotonin Mood Support
Mood – Serotonin Mood Support, Happy Saffron
Anxiety – DHA Omega-3s
Fish oil – Omega 3 Power

ACG Meds
For Worry, Anxiety and Depression – SSRIs, such as Lexapro, Paxil, Zoloft, Celexa, Prozac, and Luvox
Effexor or Cymbalta if also low PFC
Atypical antipsychotics in refractory cases

Positives w/high AC activity: Persistent, strong-willed, likes routine, consistent

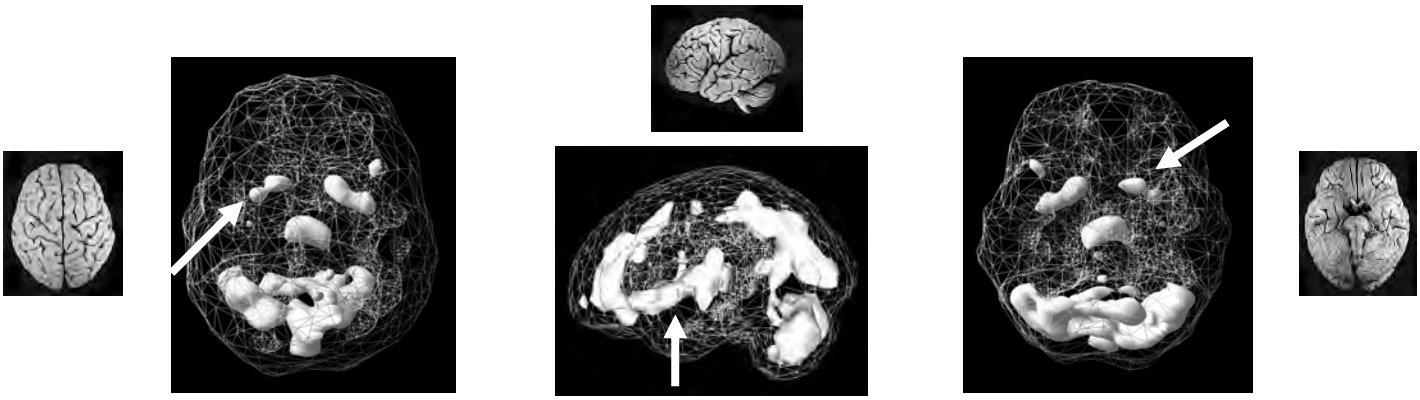
Increased AC is often associated with problems shifting attention which may be clinically manifested by cognitive inflexibility, obsessive thoughts, compulsive behaviors, excessive worrying, argumentativeness, oppositional behavior or "getting stuck" on certain thoughts or actions. We have seen a strong association with this finding and obsessive-compulsive disorders, oppositional defiant disorders, eating disorders, addictive disorders, anxiety disorders, Tourette's syndrome and chronic pain (especially when combined with increased basal ganglia activity). If clinically indicated, hyperactivity in this part of the brain may be helped by anti-obsessive antidepressants that increase serotonin. Certain forms of behavior modification techniques have also been found to help lessen activity in this part of the brain. When this area is low in activity it is often associated with low motivation and verbal expression. Clinical correlation is always needed.

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Basal Ganglia System (BGS)



top down active view | left side active view | underside active view

BG Functions **BG Problems**

Caudate

Sense of calm	Tension, nervousness
Sets anxiety level	Anxiety/panic
Conflict avoidance	Predicting the worst

Putamen

Motor related	Tremors/tics

Nucleus Accumbens

Mediates pleasure	Addiction

** ventral striatum has to do with limbic/emotional/neuroendocrine regulation, while dorsal caudate is more related to dorsal PFC/executive and 'cold' cog functions

Some Conditions Affecting the BGS

Anxiety Disorders	Tourette's/tics
OCD	PTSD
Movement disorders	

BG Behavioral Treatment

Body biofeedback	ANT therapy
Hypnosis, meditation	Relaxing music
Assertiveness training	Limit caffeine/alcohol
EMDR	

BG Supplements

Anxiety – GABA Calming, Everyday Stress Relief, Magnesium Chewables
Fish oil (Omega 3 Power)

BG Meds

For Anxiety – Buspar, anti-seizure meds, some blood pressure meds, such as propranolol may help, benzodiazepines (low dose, short time)

Positives w/high BG activity: Prepared, cautious, risk-averse, motivated

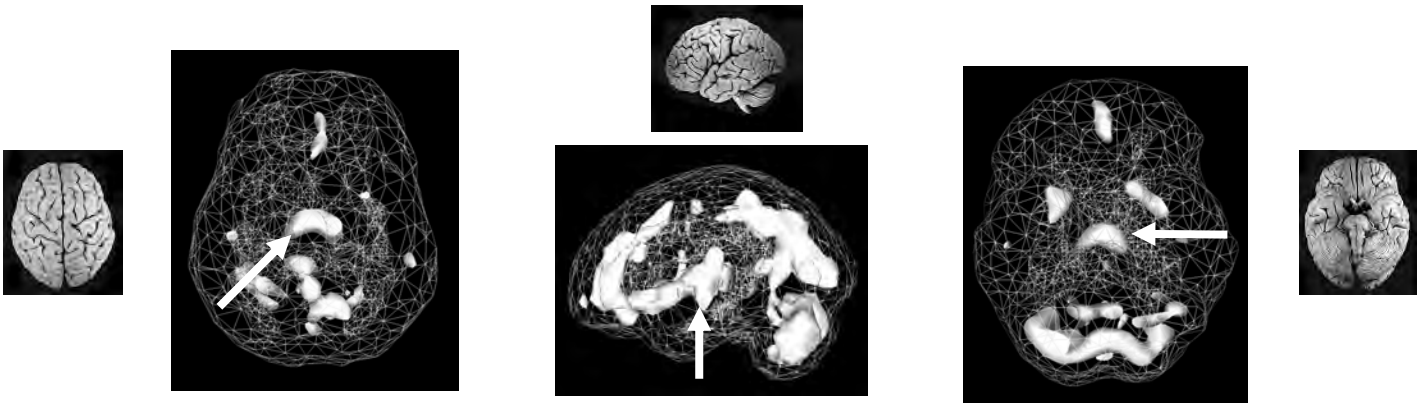
Increased basal ganglia activity is often associated with anxiety (left sided problems are often associated with irritability, right sided problems more often associated with inwardly directed anxiety). Often, we have seen increased activity in this part of the brain in our normal population as well. We have seen increased activity associated here with increased motivation. Clinical correlation is needed. We have seen relaxation therapies, such as biofeedback and hypnosis, and cognitive therapies help calm this part of the brain. If clinically indicated, too much activity here may be helped by anti-anxiety medications, such as buspirone. Sometimes, if the finding is focal in nature (more one side than the other), anticonvulsant medications can also he helpful.

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Amen Clinic Brain System Handout

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Deep Limbic System/Thalamus (DLS)



top down active view left side active view underside active view

DLS Functions
Mood control
Motivation
Attitude
Appetite/sleep
Bonding
Sense of smell
Libido

DLS Problems
Depression
Poor motivation
Poor attitude
Sleep/appetite issues
Tends to isolate
Lack of smell
Negativity, guilt
Hopelessness

Some Conditions Affecting the DLS
Depression
Cyclic mood disorders
Pain syndromes

Behavioral Treatments
Cognitive-behavioral strategies
Biofeedback, increase left prefrontal activity
Intense aerobic exercise, Relationship counseling
Increased protein diet – The Zone

DLS Supplements
Mood – Happy Saffron, SAMe, DL-phenylalanine, L-tyrosine
Fish oil (Omega 3 Power)
Cyclic Mood Changes – Fish oil, GABA Calming
For Pain - SAMe

DLS Meds
For Depression – antidepressants, such as Wellbutrin, Effexor or Cymbalta, SSRIs (if high ACG also present);
Cyclic Mood Changes -- anticonvulsants or lithium
For Pain -- Cymbalta

Positives w/high DLS activity: Sensitive, deeply feeling, empathic

Increased activity in the DLS can be associated with mood issues. Left-sided problems are often associated with anger and irritability, right sided problems more often associated with inwardly directed sadness. In our experience we have seen diffuse DLS overactivity tends to be more consistent depression and focal increased DLS activity (more on one side than the other) to be associated with cyclic mood disorders. When focal increased uptake is found in conjunction with patchy increased uptake across the cortical surface there is a higher likelihood of a cyclothymic or bipolar disorder. If clinically indicated, diffuse increased DLS uptake is often helped by antidepressant medications. If there is also increased anterior cingulate activity consider a serotonergic antidepressant. If there is not increased anterior cingulate activity consider an antidepressant which increases either dopamine (such as buprion) or norepinephrine (such as imipramine or desipramine). We use anticonvulsants or lithium to help with focal DLS hyperactivity when a cyclic mood clinical pattern is present. We have also seen increased activity in this part of the brain in our normal population. Clinical correlation is needed.

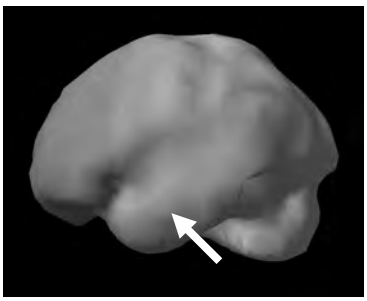
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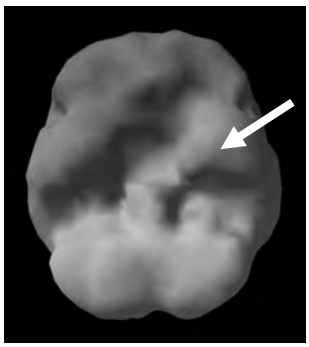
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Temporal Lobes (TLs)





left side surface view





underside surface view

TL Functions	**TL Problems**
Amygdala	Emotional valence
Emotional reactions	Mood stability
Moodiness, irritability	Temper control
Anger, anxiety, fears, Phobias, dark thoughts	
Hippocampus	
Memory	Forgetfulness
Lateral	Language
Trouble finding words	Listening
Processing problems	Reading
Poor reading	Read social cues
Poor social skills	Rhythm, music
Rhythm problems	Spiritual experience
Unusual experiences	Religiosity
	?? Sexual addiction
Inferior	Recognize faces

Some Conditions Affecting the TLs

Head injury	Dissociation
Anxiety	Temporal epilepsy
Amnesia	Serious depression

Left side – aggression, dyslexia
Right side – autistic spectrum disorders

TL Behavioral Treatments
Biofeedback to stabilize TL function
Relationship counseling, anger management
Ketogenic diet
HBOT

TL Supplements
For Mood Stability, Irritability or Anxiety – GABA Calming, valerian, fish oil (Omega 3 Power)

For Memory – Brain and Memory Power Boost (gingko and vinpocetine for blood flow, huperzine A for acetylcholine, phosphatidylserine (PS), NAC, and alpha lipoic acid

TL Meds
For Mood Stability, Irritability and Anxiety – anti-seizure medications: Lyrica, Neurontin for anxiety, irritability and pain; Lamictal and Depakote for bipolar symptoms

For Memory -- memory enhancing medications for more serious memory problems, such as Namenda, Aricept, Exelon or Reminyl

Abnormal TL (either increased or decreased) activity may be associated with mood instability, irritability, memory struggles, abnormal perceptions (auditory or visual illusions, periods of deja vu), periods of anxiety with little provocation, periods of spaciness or confusion, and unexplained headaches or abdominal pain. Left sided problems are more associated with irritability and dark thoughts, right sided more with anxiety and social struggles. Anticonvulsant medications often help with TL problems. Decreased activity in the posterior aspects of the left temporal lobes, in our experience is often, although not always, associated with language learning problems, especially reading and auditory processing problems. Memory loss is often associated with decreased activity in the medial temporal lobes.

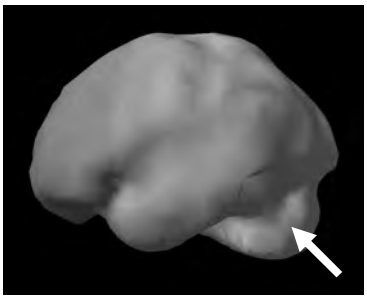
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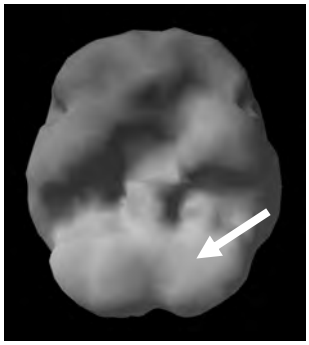
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Cerebellum (CB)





left side surface view





underside surface view

CB Functions
Motor control
Posture, gait
Executive function, connects to PFC
Speed of cognitive integration (like clock speed of computer)

CB Problems
Gait/coordination problems
Slowed thinking
Slowed speech
Impulsivity
Poor conditioned learning

Some Conditions Affecting CB
Trauma
Alcohol abuse
Autism, Asperger'sADHD

CB Treatments
Prevention of brain injury
Stop alcohol use or other toxic exposure
Occupational Therapy
Maximize brain nutrition
Hyperbaric Oxygen Therapy
Interactive Metronome
Coordination exercises such as dance or table tennis

The cerebellum is usually the most active part of the brain and is usually symmetrical in appearance. When it is low in activity it has been associated with ADD, autism, brain trauma, toxic exposure, and judgment or impulsivity issues.

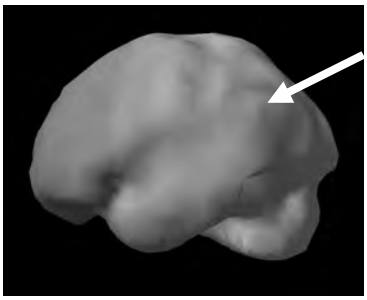
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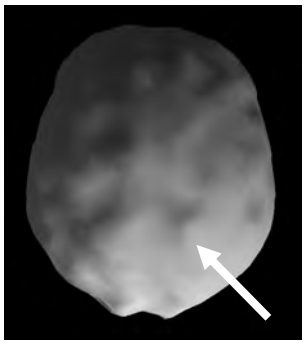
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Parietal Lobes (PLs)





left side surface view





top down surface view

PL Functions	PL Problems
Direction sense	Gets lost easily
Create and read maps	Poor map reading
Sense of touch	Oversensitive
Spatial processing	Poor spatial processing
Sees objects in space	Poor tracking
Visual guidance	Poor visual guidance
Distinguish L from R	RL Confusion
R - Personal ownership	Unilateral neglect, Construction apraxia
Admitting a problem	Denial
L – Spatial cognition	Calculation, writing, reading

Some Conditions Affecting PLs
Trauma
Toxic exposure
Infection
Anoxia
Substance abuse
Alzheimer's Disease

PL Treatments
Prevention of brain injury/toxic exposure
Occupational Therapy
Sensory Integration Therapy
Maximize brain nutrition
Hyperbaric Oxygen Therapy.

Increased activity in the PLs often indicates hypersensitivity to noise, touch and taste. The parietal lobes have also been implicated in attentional issues and distractibility (too much stimulation comes in). In addition, when there is both sided parietal lobe and frontal lobe activity it is often associated with a pattern we call the ring of fire. Decreased activity in the PLs may be associated with trauma, toxicity, or memory loss.

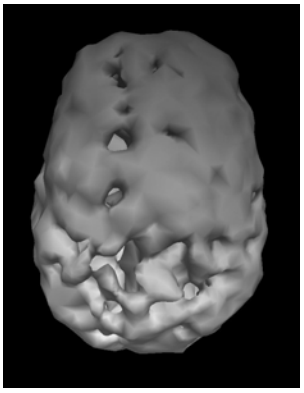
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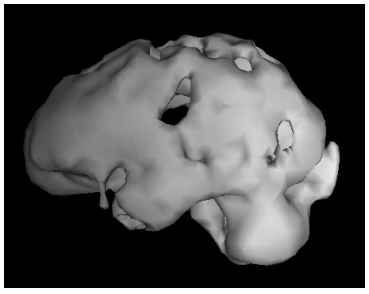
Toxicity/Scalloping



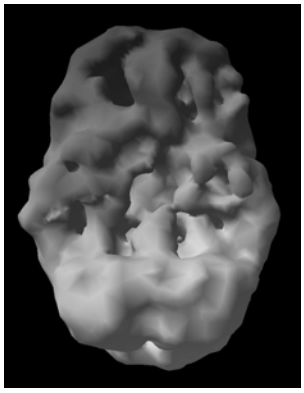


top down surface view





left side surface view



underside surface view



Toxicity/Scalloping
On scans toxicity shows up as scalloping, bumpy or irregular appearance

Toxicity/Scalloping Causes
Drug or alcohol abuse/exposure
Environmental toxic exposure (paint fumes, solvents, metals, pesticides, pollution, mold, etc.)
Oxygen deprivation
Brain infections
Cancer chemotherapy
Anesthesia/surgery
Metabolic issues (thyroid, liver, kidney)

Toxicity/Scalloping Symptoms (depends on system affected)
Cognitive problems (attention and memory)
Fatigue
Irritability
Prefrontal, temporal, and/or parietal lobe symptoms

Toxicity/Scalloping Treatments
Eliminate offending agent(s)
Maximize brain nutrition
Hyperbaric Oxygen Therapy
Brain healthy strategies to consider:

- Both physical & mental exercise
- Avoid behaviors risky for brain injury
- Omega-3 fatty acids
- Alpha lipoic Acid, 100-300mg 2XD
- Multiple vitamin
- Vitamin E 100 IUs 2XD and Vitamin C 1,000mg 2XD as antioxidants
- Brain and Memory Power Boost 3 2XD
- Coenzyme Q10 (CoQ10) help with energy/ memory, especially if Parkinson's disease is present or in family. 100 – 400 mg/day

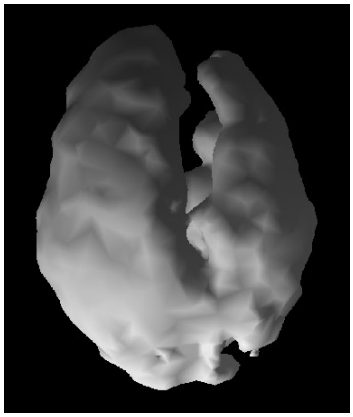
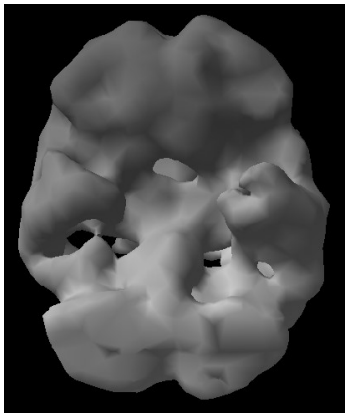
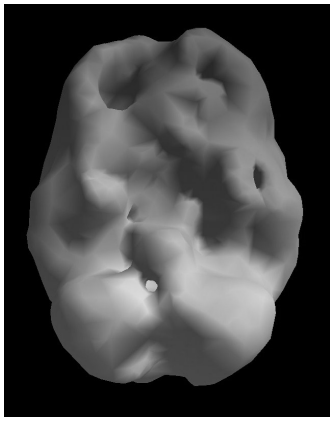
Toxicity or scalloping is often seen on scans and may be due to a number of causes, such as substance abuse, environmental toxins, oxygen deprivation, infections, etc. Scans cannot date the time of exposure, but say if there is increased likelihood that they occurred. When toxicity is present it is critical to eliminate the offending agent and work to regain the best brain health possible.

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Amen Clinic Brain System Handout

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Physical Brain Trauma Patterns


top down surface view | underside surface view | underside surface view

Physical Brain Trauma Patterns

Brain trauma can show in many different ways on scans. Typically, there is:

- Focal areas of decreased activity in the prefrontal cortex, temporal lobes, parietal lobes, occipital lobes
- Decreased prefrontal pole
- Decreased anterior temporal poles
- Decreased temporal lobes
- Focal areas of increased activity
- Off center anterior cingulate activity

Trauma Causes

Blunt force
Whiplash injuries
Acceleration/deceleration injuries
Falls, sports, motor vehicle accidents
** loss of consciousness may not be present

Trauma Symptoms (depends on system affected)

Cognitive problems (attention and memory)
Fatigue
Irritability
Prefrontal, temporal, parietal lobe symptoms

Physical Brain Trauma Treatments

Eliminate risk for future injuries
Maximize brain nutrition
Hyperbaric Oxygen Therapy
Neurofeedback over damaged area
Brain healthy strategies to consider:

- Both physical & mental exercise
- Avoid behaviors risky for brain injury
- Omega-3 fatty acids
- Brain and Memory Power Boost 3 2XD
- Multiple vitamin
- Coenzyme Q10 (CoQ10) help with energy/ memory, especially if Parkinson's disease is present or in family. 100 – 400 mg/day

Physical brain trauma is more common than people realize. We often see damage on scans of what most people would consider minor brain trauma. Scans cannot date the time of the trauma, but say if there is increased likelihood that they occurred. When trauma is present it is critical to eliminate future risk and work to regain the best brain health possible.

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